

Animation system: Animations define high-level transformations, not frame-by-frame changes.

General animations are:

- `Create()` → Object creation animation
- `Transform()` → Morph one object into another
- `FadeIn()` / `FadeOut()` → Opacity transitions
- `Write()` → Text-specific writing animation

Timing control is as follows:

- `self.play()` → Executes animation sequences
- `self.wait()` → Adds pause or delay
- `run_time` → Controls animation speed

Setting up Manim on Linux

To install and run Manim, we need Python 3.7 or higher, FFmpeg (for video encoding), LaTeX (for mathematical typesetting) and Cairo (for graphics rendering).

Let us start with dependency installation.

For Ubuntu/Debian:

```
sudo apt update
sudo apt install python3 python3-pip python3-venv
sudo apt install ffmpeg sox libcairo2-dev pkg-config
sudo apt install texlive-full pandoc
```

For Arch Linux:

```
sudo pacman -S python python-pip ffmpeg
sudo pacman -S texlive-most cairo
```

For Fedora:

```
sudo dnf install python3 python3-pip ffmpeg
sudo dnf install texlive cairo-devel
```

For a virtual environment setup, create an isolated environment:

```
python3 -m venv manim-env
source manim-env/bin/activate
pip install --upgrade pip setuptools wheel
```

To install Manim, type:

```
pip install manim
```

For installation verification, use the following code:

```
manim --version
manim checkhealth
```

Common solutions are:

- Missing LaTeX → Install texlive-full package
 - Cairo errors → Install libcairo2-dev development headers
 - FFmpeg issues → Verify PATH and installation
- For directory structure setup, organise project directories:

```
mkdir -p ~/manim-projects/{scripts,media,assets}
cd ~/manim-projects/scripts
```

Creating your first animation

For basic scene creation, create a file *first_animation.py*:

```
from manim import *

class BasicShapes(Scene):
    def construct(self):
        # Object creation
        welcome_text = Text("Welcome to Manim!")
        square = Square(side_length=2, color=BLUE)
        circle = Circle(radius=1.2, color=GREEN)

        # Positioning
        welcome_text.to_edge(UP, buff=0.5)
        square.shift(LEFT * 2.5)
        circle.shift(RIGHT * 2.5)

        # Animation sequence
        self.play(Write(welcome_text))
        self.play(Create(square), Create(circle))
        self.play(square.animate.rotate(PI/4))
        self.play(circle.animate.set_color(YELLOW))
        self.play(Transform(square, circle))
        self.wait(2)
```

The output of the above code can be seen Figure 2.

For basic rendering, use:

```
manim -pql first_animation.py BasicShapes
```

The flag meanings are:

- p -- Preview (auto-open video)
- q -- Quality (l=low, m=medium, h=high)



Figure 2: Basic scene creation